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Performance of Disabled Persons on a Chordic Keyboard

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Fourteen disabled persons used a one-hand chordic device for typing. This keyboard was designed to minimize physical exertion and to inhibit unwanted psychomotor reactions so as to facilitate its use by persons with cerebral palsy, muscular dystrophy, and related disorders. The keyboard fits the fingers of one hand. A character is typed by pressing a combination of fingers corresponding to a typing code developed earlier. Typing rates of text transcription ranged from 8 to 14 words/min after 5 h of practice. These results indicated that the physical configuration and cognitive operation of a chordic keyboard would permit disabled persons to use computers.

INTRODUCTION

The widespread dissemination of computer technology into work organizations, leisure activities, and educational institutions is bypassing the disabled. This trend has been widely documented and highlighted in reports issued by the Office of Technology Assessment (1982), the National Academy of Science (1976), and the U.S. Department of Education (1982). According to Sheridan (1978) and Vanderheiden (1982), this is due primarily to the lack of an appropriate interface unit to adapt the computer keyboard to the needs of the disabled. Recent efforts at creating such an interfacing device have been restricted in scope and targeted for an extremely small population, mainly those who are sensory disabled (Kirschenbaum, 1984). Most efforts involve, through physical and/or software alterations, a modification of the existing standard 80 to 90-key keyboard (Shoefield, 1981). Nearly all are based on a single-switch operation with character selection involving a scanning procedure of an alphanumeric matrix. Our study evolved in response to the need for an interfacing unit that would help to bridge this gap.

KEYBOARD DESIGN

The use of standard keyboards to operate computers by persons with motor, sensory, or mental disabilities is extremely difficult. According to Siebel (1972) and Kroemer (1972), the standard keyboard, composed of about 80 closely packed nonlogically distributed keys, requires physical dexterity and complex psychomotor sequences linked to hand-finger-eye coordination. This presents a formidable physical and psychological barrier for disabled persons. An alternative interface unit should specifically consider the

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need for an appropriate physical configuration suitable to a variety of disabilities. In addition, it should provide an easy way of learning and should be adjusted to the requirements of the end user. These criteria, which guided the initial stages of our study, led us to use a one-hand chordic keyboard. Pioneering attempts to use such keyboards in a more general setup have been initiated by Conrad and Longman (1965) and Gopher and Eilam (1979), and later by Gopher and Koenig (1983).

A simple modification of the standard keyboard cannot fulfill the demand for a flexible and effective interface keyboard for the disabled. Careful analysis of the characteristics of major disability groups pointed toward the need for a flexible physical configuration that would require minimum physical exertion. This is attested to in the latest analysis of the characteristics of the disabled, found in the publication of the Office of Technology Assessment (1982). Detailed reviews of associated problems are provided in LaRocco and Turem (1978) and Brown and Redden (1979).

To this end, the design concentrated on a hand-extension, usable by one hand but easily adaptable for the disabled with partial use of either one or two hands and encompassing a learning procedure readily absorbed and effective as a motivating device. In addition, consideration was given to aesthetics, smallness, and mobility. These requirements were deemed essential for acceptance by the disabled, thereby having potential impact on the success of the computer interface unit.

The chosen interface unit is essentially a mold extension of the hand. Its actual design, based on the work of Eilam (1982), closely matches the physiology of the hand and movements of the fingers. We call this chordic keyboard *helpwrite*. It is illustrated in Figure 1.

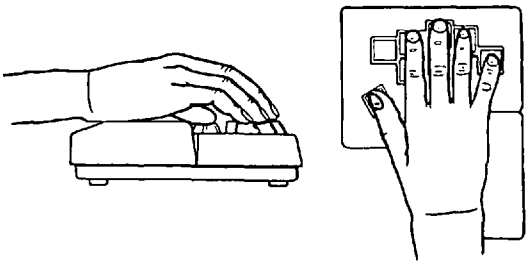

Figure 1. *Illustration of chordic keyboard used by persons with disabilities.*

Preliminary experiments on prototype models indicated a one-hand extension design to be beneficial for disabled persons with reduced dexterity of the arm-hand-finger limb movement (e.g., persons with muscular dystrophy, spasticity, arthritis, or paralysis). The "final" design, as illustrated in Figure 1, consists of five major key positions and three shift-control keys arranged to match the natural spread of the hands and fingers. The control keys are lined up with the character keys of the strongest fingers. This design takes into account variations in hand size, length of fingers and muscular exertion patterns. An example of this is the placement of the pinky key (slightly higher than the other keys). This finger positioning improves performance. In addition, the physical configuration of the keys (which are larger and easily accommodate the size and curvature of fingertips) assists disabled persons with poor motor control of the hand. A final factor, which proved essential, is the stable position of the hand and fingers, minimizing movement from a fixed position on the keyboard.

The basic keyboard carries a definite potential for accurate and speedy transcription. It should be noted that when the operator rests the hand on the keyboard interface itself, the only physical operational require-

ment involves the fingers. For many types of disabilities, this residual ability remains despite the loss of control of other physical functions.

CHORDIC PRINCIPLE

Most disabled persons have little or no previous experience in typing, but this should have little impact on their learning abilities. Previous experiments by Gopher (1984) on nondisabled persons, with and without typing skills, indicated little difference in learning time, speed, or accuracy.

Most languages have characters with high probability of use. This also includes specific combinations of letters. An analysis of the physiology of the hand and fingers revealed a similar high frequency of use of certain fingers. Combining these two factors led to an alternative number of alphanumeric codes, in which repetitive characters are linked to fingers used with high frequency. A series of experiments conducted by Gopher and Eilam (1979) led to a code, which takes advantage of the "natural" features of hand-finger movements and of natural language. An example is the letter *e* linked to the index finger, or the letter *r* linked to the combination of the thumb and middle finger together. Figure 2 illustrates the basic letter combinations.

For the disabled, such ease of operation, in which minimum effort is required and where human capabilities are used in the most natural manner, seems ideal as an interface replacing the standard keyboard.

A second aspect concerns the actual learning process. Here, a simple learning technique should be employed in order to minimize time and effort. Again, the disabled, due to their impairments, would require a best-fit means of acquisition, retention, and recalling such a code. Without easy learning, a combination of physical and psy-

	DIGITS				
	THUMB	FOREFINGER	MIDDLE	RING	PINKY
A		•	•		
B	•	•	•	•	•
C	•	•	•		
D	•				•
E		•			
F	•		•	•	•
G		•	•	•	•
H				•	•
I	•			•	
J	•	•		•	•
K	•	•	•	•	
L	•	•			
M	•		•	•	
N					•
O				•	
P			•	•	•
Q	•			•	•
R	•		•		
S			•	•	
T			•		
U		•	•	•	
V		•			•
W	•	•		•	
X	•	•			•
Y		•		•	
Z		•		•	•
SPACE	•				

Figure 2. *Alphanumeric typing code indicating finger-key combinations needed to elicit single letters.*

chological barriers might discourage full utilization of the interfacing device. Therefore, a mnemonic learning principle is used. It is based on catch words or phrases, which are then linked with certain combinations of keys. For example, the depression and release of each key singly is linked with the term *NOTE-space*. From right to left, depression of a single key creates a single letter: the first, *N*; the second, *O*; the third, *T*, and so on. Another example is the mnemonic aid, *CUP*, in which three adjacent keys are depressed simultaneously. Again, it should be recalled that letter and finger are matched by repetition and frequency of use and not by alphabetical order.

METHOD

In the latter part of 1983, a special course was introduced to teach disabled persons how to use the computer. The course was held in a barrier-free environment that was appropriate to the needs of disabled persons. Registration was open to anyone, with the training sessions held three times per week. Prior to the course, registrants were interviewed with regard to their biographical and medical histories, and their capabilities were assessed, particularly hand coordination and cognitive ability.

The training sessions were built around a training manual. The manual was devised on the basis of pretesting both disabled and nondisabled persons. This pretesting allowed us to create a learning program schedule, which focused on an accumulation of alphanumeric characters in their mnemonic order of complexity. Thus, letters based on a single key depression were followed by double key (LASH), triple key (CUP), and so on. Each lesson included the new set of letters and a repetition of previously learned characters. In this way, every success brought about the reward of additional letters and the ability to type more complex words and sentences. By learning 15 letters in this series, the student was able to construct close to 75% of most-used words in a language.

Fourteen persons with varying disabilities were selected for the experiment. Our objective was to determine how successful our approach would be in ameliorating barriers for computer interaction. In particular, recording of learning time and accuracy of retention of the code—and its successful representation through typing—would allow a reasonable test of the interface keyboard for such disabled persons.

The sample itself contained persons ranging in age from 8 to 25 years; including individuals still in elementary school and graduates potentially in the labor market. Included clinical disabilities were cerebral palsy (athetoid and spastic), muscular dystrophy, and dyslexia. The key physical features were related to arm-hand-finger control. Here, most of the sample did not have good motor control of the hand/fingers. In fact the majority had to use two hands to operate the interface keyboard, manipulating fingers and hand positions to use the keys.

The ability to match the symbolic code, its representative key and appearance on the monitor, also varied. Discriminating between a correct and mistaken output of the keyboard onto the monitor affected accuracy and speed of learning, as will be noted later. Its importance lies in the fact that a large proportion of physically disabled persons have concomitant cognitive problems (either directly related to their disability or through environmental factors). It should also be pointed out that each student had a 30-min lesson three times per week. No student was given the keyboard interface between these lessons for practicing. The learning schedule was similar for all students. They set their own pace and were tested both at the end of each lesson and prior to the next.

Each student was introduced to the key-

board and training station through a short introduction of how to physically operate the keyboard. The student was then given about 30 min to explore the keyboard and find a best-fit position in which to be able to fully utilize all alphanumeric characters through key combinations. In some cases, this led to use of fingers on two hands (the most common was the thumb of one hand and four fingers on the other) or, in cases of severe physical constraints, to a shifting and repositioning of the keyboard to match restrictions of hand movement.

RESULTS

After the introductory lesson, the students were given ten 30-min lessons. None of the students dropped out of the program. Each lesson, as was noted, was cumulative and moved from simple to more complex combinations to attain the desired alphanumeric characters. The data in Figure 3 illustrate the continuous increase in typing proficiency with each session. All of the students mastered the entire alphanumeric code (letters and numbers) by the end of the eleventh lesson, with each progressive session reflecting past and newly acquired codes.

The majority of those in the experiment were able to correctly type between 2 and 4 word/min by the end of the first session. By the last lesson, typing skill had improved to an average of 12 word/min. It should be noted that these figures represent correct typing of free-floating text. Mistakes had to be corrected; only clean text was acceptable.

Figure 4 illustrates the average time the students required to master each mnemonic code—which progressed from single key to more complex combinations—by correctly recalling and typing it and previous codes as the learning sessions continued. The data reveal that learning and recalling cumulative codes takes less time as the sessions progress. We had expected that, due to the complexity of the code combinations and necessity to recall all of the previous codes, the time needed to type correctly would increase (i.e., decrease proficiency and speed). However, once the first three groups of combinations was learned, there was a decline in time required to learn additional code combinations. Apparently, the move from simple single-stroke operation to the first set of combinations (the second lesson) requires a skill that increases

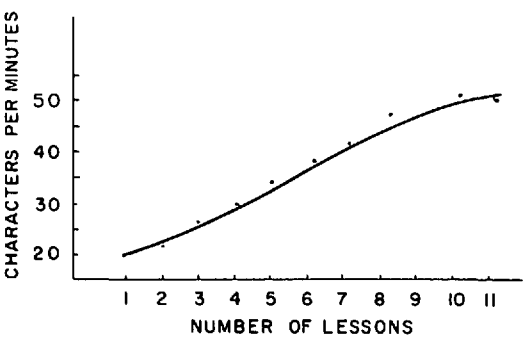

Figure 3. *Average number of characters typed per minute by number of lessons each student participated in.*

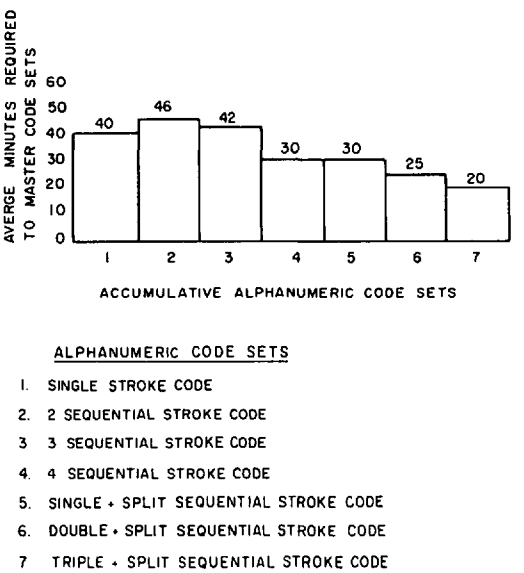

Figure 4. *Average time to master accumulative alphanumeric code sets.*

learning time. Once this is gained, even more complex combinations of successive keystrokes pose little problem. Overall, the time needed to acquire the codes is halved despite the fact that the end codes are cumulative and more complex than at the beginning.

Among physically constrained subjects, there is a considerable variability in typing proficiency. This is due to the characteristics of the given disability that affect physical and cognitive abilities. For example, the average number of words per minute typed by spastic cerebral palsy subjects is lowest of all disability groups studied (eight words/min). A distinctive characteristic displayed by members of this disability group is tense and contracted muscles, often leading to a lack of control and coordination. Such individuals are typically unable to produce fine hand/finger movements, and they therefore have difficulty in coordinating their movements to operate the one-hand keyboard. The lack of coordinated movement among spastic individuals is usually exacerbated by the increased effort required to perform an activity. The keyboard, which was designed to minimize such effort apparently has a positive effect on the degree of coordination that can be attained by persons with such a disability.

For persons with athetoid cerebral palsy, involuntary and unorganized movements are typical. This is particularly true for gross motor movements which, in turn, affect finer hand-finger coordination. Here again, the keyboard requires practically no movement of the arm or hand, which may lead to decreased involuntary activity. The fact that this disability group averages about 13 words/min clearly indicates that the keyboard interface unit is appropriate.

A third disability group is progressive muscular dystrophy (PMD). Here, the muscular system of the body deteriorates, radically reducing the ability to activate or coordinate extremities and eventually vital organs. Increasing weakness slows down movement and restricts gross motor movement of the arm and hands. Yet, due to the need for minimum physical effort to operate the interface keyboard, persons with PMD can type successfully. The 14 word/min average at the end of the eleventh session attests to this.

Emphasis on the hand-extension quality of the interface led us to further investigate the effect of hand control over typing proficiency. The data clearly suggest that hand control is a key factor in proficient typing. Persons with good hand control, in contrast to those with poor control, have rapid rates of transcription. These results matched the typical characteristics of the disability groups examined. Importantly, even those with poor hand control (primarily with severe spastic cerebral palsy) were able to learn the entire alphanumeric code and type accurately at a slow rate. Typing on a standard keyboard would be impossible for such individuals.

A more detailed analysis of the varying positions of hand control among subjects with fair to poor control is particularly interesting. It points out the degree of flexibility of the interface unit and the adaptability of the disabled in taking advantage of it. An extreme example is a male teenager who has absolutely no control over fingers in both hands. Yet, by repositioning the interface unit and employing the edge of his hand, he was able to accurately type more than 10 words/min at the end of 10 sessions. Another case involves a woman in her twenties who is severely spastic and uses both hands—combining finger with gross hand movements—to operate the keyboard.

In the case of subjects with extremely weak hands (PMD), two variations appeared in the use of the keyboard: a full one-hand use or two-hand use due to extreme weakness of

specific fingers on each hand. In both cases, the ability to place the keyboard in a comfortable position allowed its optimal use. In addition, even when there was little or no control in arm and hand functions, enough residual muscle control was available for complete operation of all alphanumeric combinations.

A difficult issue involves the possible contribution of cognitive ability. The nature of our sample precludes us from reaching a definite conclusion about this. However, casual observation points out that subjects with good cognitive ability learn faster than those with average or poor cognitive ability. It should be noted, however, that the average proficiency at the end of the experiment does not tell us about the subjects' ability to learn the mnemonic code nor whether their proficiency increased at a steady or erratic rate over time.

CONCLUSIONS

The objective of this study was to evaluate the use of a chordic keyboard by the disabled. Previously, interface units for the disabled have concentrated on a small number of severely disabled persons. Such units are constricted in use and conceived for a specific problem, usually for a single individual. In this study, however, our aim has been to seek a general solution covering a broad range of disability groups. A preliminary analysis of the characteristics of disability categories pointed toward the hand/fingers as a predominant ability area common to many types of impairments. Utilizing this commonality, which encompasses persons with motor, sensory, and mental impairments, led to the use of a single hand-extension interface keyboard.

The initial results strongly suggest that our major objective has been fulfilled. Disabled persons with various impairments are now able to successfully operate a keyboard interface unit and consequently to interact with computers. Admittedly, these results do not substitute for large-scale evaluation programs. The chordic keyboard proved flexible and helpful to the disability categories examined. Its physical design and cognitive operation allow persons who could not type to interact with computers.

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